

Remove and Install Countershaft Replacement - Stark VARG MX 1.2 / EX

Video link: https://www.youtube.com/watch?v=N5_INHPwDQ0

Source video: Remove and Install Countershaft Replacement - Stark VARG MX 1.2 by Stark Future Official

Applicable models: MX 1.2 / EX

Scope

This guide converts the official Stark Future tutorial video into a step-by-step workshop procedure. It documents each action shown in the video in sequence and pairs every step with a dedicated screenshot.

Preparation

- Place the bike on a stable stand or work area before starting the procedure.
- Prepare the correct tools for the fasteners, clips, brackets, and components shown in the video.
- Keep removed hardware organized in sequence so reassembly follows the same order as the video.
- Have a torque wrench ready for any tightening steps that specify a torque value.

Removal Procedure

1. Remove the four skid plate screws

Remove the four skid plate screws



2. Carefully remove the skid plate from the battery

Carefully remove the skid plate from the battery.



3. Unscrew the oil drain plug

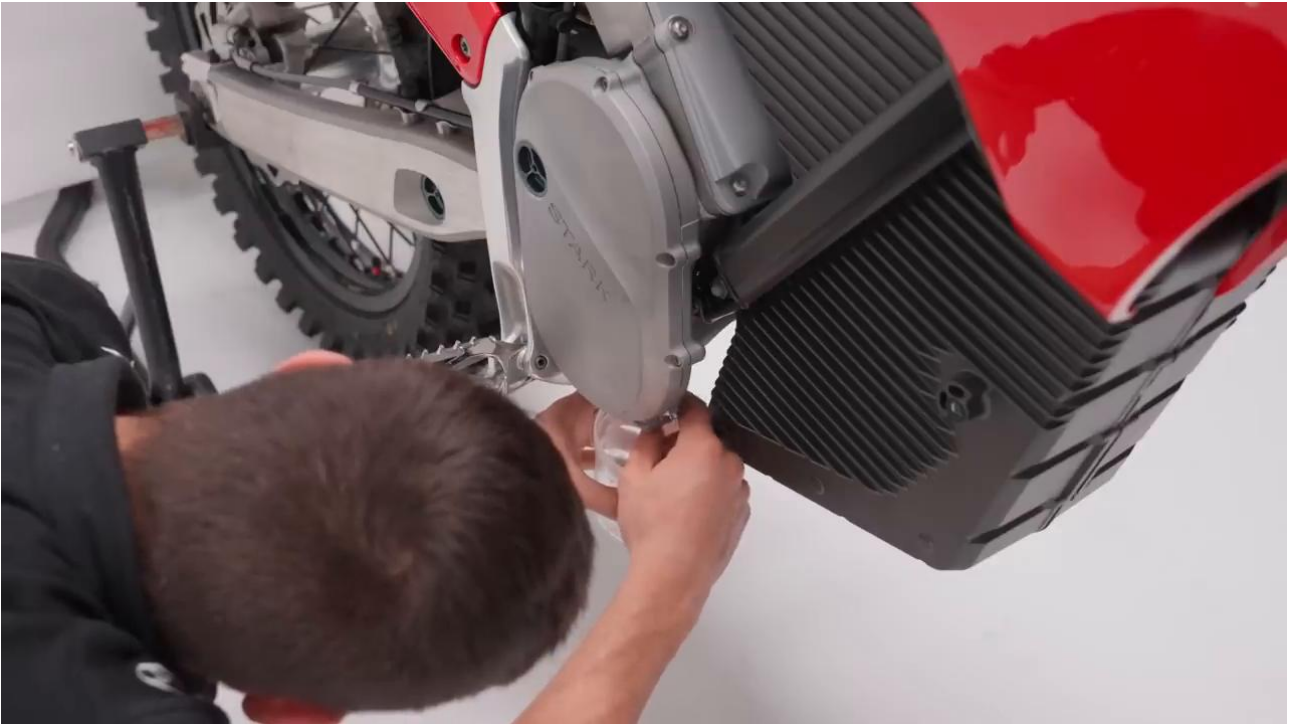
Unscrew the oil drain plug.

Carefully remove and dispose the copper washer.



4. Place a container underneath to collect the used oil

Place a container underneath to collect the used oil.



5. Unscrew the motor case side cap

Unscrew the motor case side cap.



6. Remove the eight bolts securing the right motor cover

Remove the eight bolts securing the right motor cover.



7. Carefully remove the right motor cover

Carefully remove the right motor cover.

Carefully remove and dispose of the gasket.



8. Prevent the motor main shaft gear and the countershaft gear from rotating

Prevent the motor main shaft gear and the countershaft gear from rotating.

A properly placed piece of cloth can be used to gently block them.



9. Remove the front sprocket guard

Remove the front sprocket guard.



10. Loosen the front sprocket bolt while applying the rear brake

Loosen the front sprocket bolt while applying the rear brake.

If necessary, use additional assistance to secure the rear wheel.



11. Carefully remove the front sprocket

Carefully remove the front sprocket.



12. Carefully open the countershaft retaining ring

Carefully open the countershaft retaining ring.



13. Loosen and remove the countershaft nut

Loosen and remove the countershaft nut.



14. Loosen and remove the motor core gear bolt

Loosen and remove the motor core gear bolt.



15. Carefully remove the main shaft gear

Carefully remove the main shaft gear.



16. Carefully remove the countershaft gear

Carefully remove the countershaft gear.



17. Before removing it, loosen and remove the three bolts securing the motor bearing retainer

Before removing it, loosen and remove the three bolts securing the motor bearing retainer.



18. Remove the countershaft spacer bushing

Remove the countershaft spacer bushing.



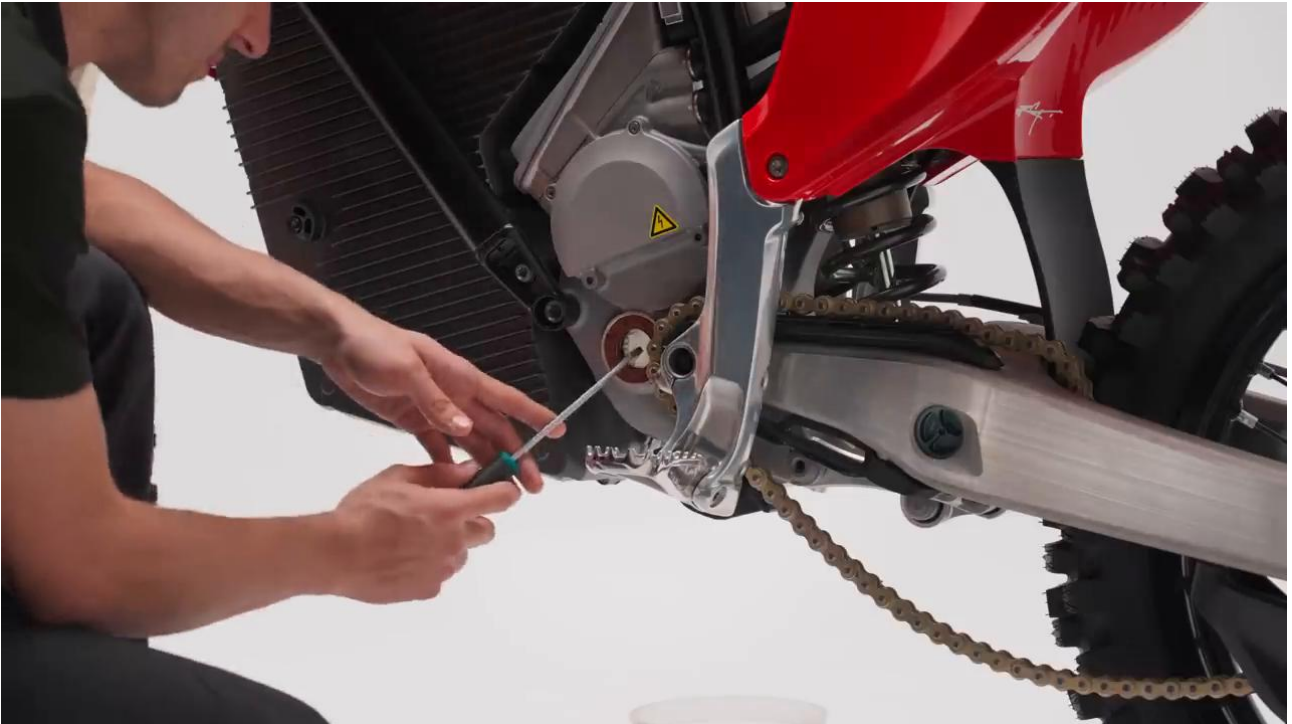
19. Using a rubber mallet, gently tap the countershaft and bearings outward to remove them from the motor case

Using a rubber mallet, gently tap the countershaft and bearings outward to remove them from the motor case.



20. Remove the countershaft seal

Remove the countershaft seal.



21. Using a pin punch, carefully remove the roller bearing from the motor case

Using a pin punch, carefully remove the roller bearing from the motor case.



22. Thoroughly clean all housing surfaces

Thoroughly clean all housing surfaces.



Installation Procedure

23. Using a pin punch, carefully install the roller bearing into its correct position

Using a pin punch, carefully install the roller bearing into its correct position.

Ensure the bearing is oriented properly.



24. Place the ball bearings into the countershaft

Place the ball bearings into the countershaft.

Ensure the bearings are oriented properly.



25. Install the circlip onto the countershaft to secure the ball bearing in position

Install the circlip onto the countershaft to secure the ball bearing in position.



26. Install the countershaft spacer bushing onto the countershaft

Install the countershaft spacer bushing onto the countershaft.

Ensure the bushing is oriented correctly.



27. Using a heat gun, gently warm the housing area to facilitate installation of the ball bearings

Using a heat gun, gently warm the housing area to facilitate installation of the ball bearings.

Do not apply excessive heat.



28. Insert the countershaft assembly into the motor case housing

Insert the countershaft assembly into the motor case housing.



29. Install the motor bearing retainer and secure it by tightening the three retaining bolts to 5 Nm using thread locker

Install the motor bearing retainer and secure it by tightening the three retaining bolts to **5 Nm** using thread locker.



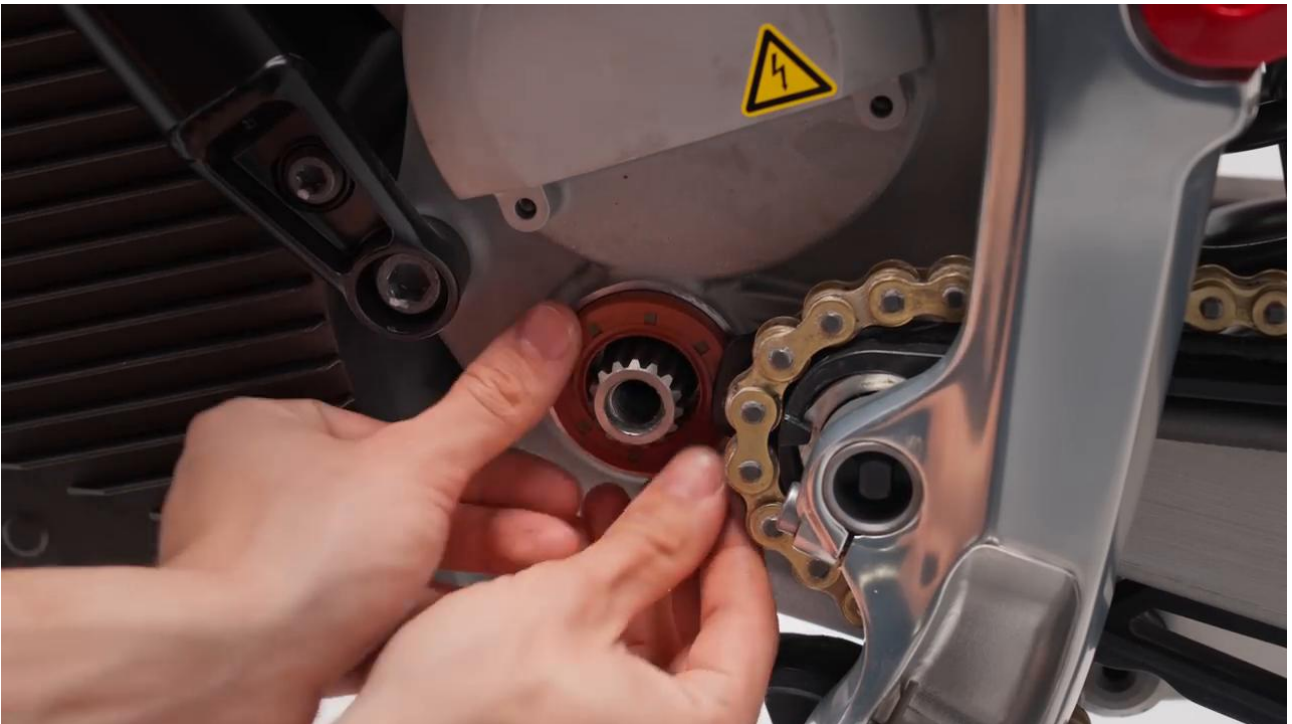
30. After cleaning the motor core gear bolt and installing the main shaft gear, prevent it from rotating against the countershaft gear and tighten the bolt to 80 Nm using thread locker

After cleaning the motor core gear bolt and installing the main shaft gear, prevent it from rotating against the countershaft gear and tighten the bolt to **80 Nm** using thread locker.



31. Install the countershaft seal all the way into its proper position

Install the countershaft seal all the way into its proper position.



32. Install the O-ring onto the countershaft

Install the O-ring onto the countershaft.



33. Install the countershaft spacer bushing

Install the countershaft spacer bushing.



34. Carefully install the front sprocket

Carefully install the front sprocket.

Ensure it is positioned correctly.



35. Carefully apply thread locker to the front sprocket bolt and tighten it to 80 Nm while applying the rear brake

Carefully apply thread locker to the front sprocket bolt and tighten it to **80 Nm** while applying the rear brake.

Ensure the washer is correctly positioned on the bushing.



36. Install the front sprocket guard by tightening the two bolts to 5 Nm

Install the front sprocket guard by tightening the two bolts to **5 Nm**.



37. Install the right motor cover and gasket, then tighten the eight bolts to 5 Nm

Install the right motor cover and gasket, then tighten the eight bolts to **5 Nm**.



38. Place a copper washer on the oil drain plug, then tighten it to 15 Nm

Place a copper washer on the oil drain plug, then tighten it to **15 Nm**.



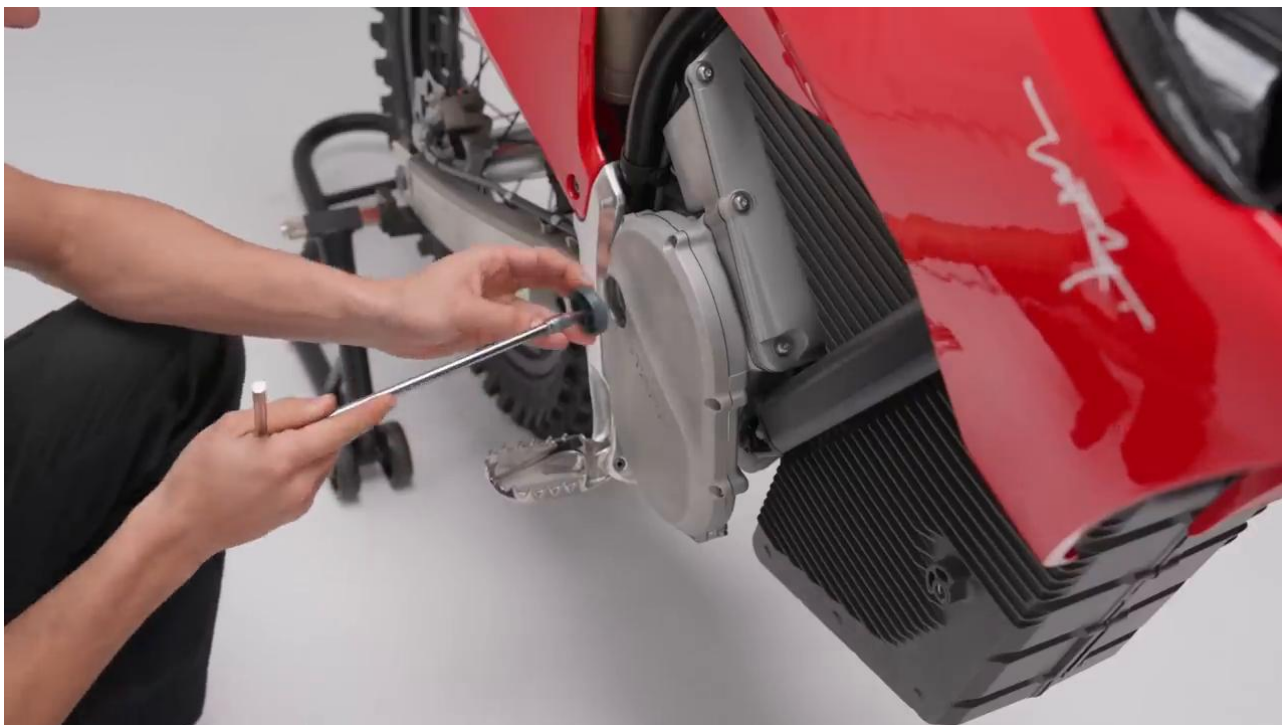
39. Fill the transmission with 80 ml of SAE 75W-90 oil

Fill the transmission with 80 ml of SAE 75W-90 oil.



40. Tighten the motor case side cap to 5 Nm

Tighten the motor case side cap to **5 Nm**.



41. Place the skid plate into its position

Place the skid plate into its position.



42. Tighten the four skid plate bolts to 15 Nm

Tighten the four skid plate bolts to **15 Nm**.



Torque Summary

Component	Torque
Install the motor bearing retainer and secure it by tightening the three retaining bolts	5 Nm
After cleaning the motor core gear bolt and installing the main shaft gear, prevent it from rotating against the countershaft gear and tighten the bolt	80 Nm
Carefully apply thread locker to the front sprocket bolt and tighten it	80 Nm
Install the front sprocket guard by tightening the two bolts	5 Nm
Install the right motor cover and gasket, then tighten the eight bolts	5 Nm
Place a copper washer on the oil drain plug, then tighten it	15 Nm
Tighten the motor case side cap	5 Nm
Tighten the four skid plate bolts	15 Nm

Video-sourced torque items:

- Install the motor bearing retainer and secure it by tightening the three retaining bolts: **5 Nm** from the official Stark tutorial video.
- After cleaning the motor core gear bolt and installing the main shaft gear, prevent it from rotating against the countershaft gear and tighten the bolt: **80 Nm** from the official Stark tutorial video.
- Carefully apply thread locker to the front sprocket bolt and tighten it: **80 Nm** from the official Stark tutorial video.
- Install the front sprocket guard by tightening the two bolts: **5 Nm** from the official Stark tutorial video.
- Install the right motor cover and gasket, then tighten the eight bolts: **5 Nm** from the official Stark tutorial video.
- Place a copper washer on the oil drain plug, then tighten it: **15 Nm** from the official Stark tutorial video.
- Tighten the motor case side cap: **5 Nm** from the official Stark tutorial video.
- Tighten the four skid plate bolts: **15 Nm** from the official Stark tutorial video.

Critical Checks Before Riding

- Confirm the serviced component is fully seated and all fasteners are tightened in the correct order.
- Recheck all torque-critical hardware before riding.
- Make sure all routed cables, clips, brackets, and surrounding parts are returned to their original positions.
- Inspect the bike visually for any missing hardware before returning it to service.